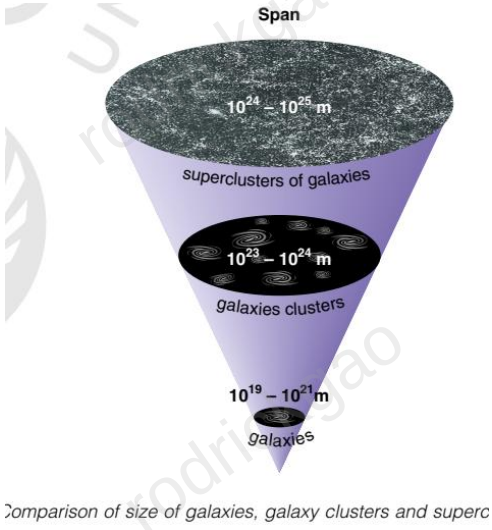
E101 Introduction to Astronomy

E1.1.1 Celestial structures and their sizes

Celestial structures include stars and planetary systems, star clusters, galaxies, galaxy groups or clusters, and superclusters. A star cluster is a gravitationally bound group of stars; a nebula is interstellar gas and dust; a galaxy contains many stars, clusters and nebulae. A star cluster is not a galaxy cluster. Identify the type of object and use consistent length units when comparing sizes, whose ranges may overlap.



Comparison of size of galaxies, galaxy clusters and superc

Compare the size hierarchy of galaxies, galaxy clusters and superclusters.

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E1.1.2 The astronomical unit

The astronomical unit, AU, is a length unit suited to Solar System distances, approximately the mean Earth-Sun distance or 1.50×10^{11} m. Convert AU, metres and other units consistently before comparing distances. An AU is neither Earth's orbital circumference nor a fixed Earth-to-planet separation.

$$1\ \text{AU} \approx 1.50 \times 10^{11}\ \text{m}$$

One astronomical unit is approximately the mean Earth–Sun distance. Textbook rounded conversion for Solar System distances.

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E1.1.3 · E1.1.4 The light-year

A light-year is the distance light travels in vacuum in one year, approximately 9.46×10^{15} m. It is a length, not a time. Calculate it from light speed times the number of seconds in a year. A star's distance in light-years indicates approximate light-travel time in years, not its age.

$$1\ \text{ly} \approx 9.46 \times 10^{15}\ \text{m}$$

A light-year is the distance travelled by light in vacuum in one year; a unit of distance, not time.

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E1.1.5 The geometrical meaning of a parsec

A parsec is the distance corresponding to an annual parallax of one arcsecond. The geometrical baseline is 1 AU, giving approximately 206265 AU or 3.26 light-years by the small-angle relation. An arcsecond measures angle, not time; a parsec measures distance despite the historical reference to seconds in its name.

$$1\ \text{pc} \approx 3.09 \times 10^{16}\ \text{m} \approx 3.26\ \text{ly}$$

$$1\ \text{pc} \approx 206\,265\ \text{AU}$$

Parsec: distance corresponding to annual parallax of one arcsecond. Rounded conversions.

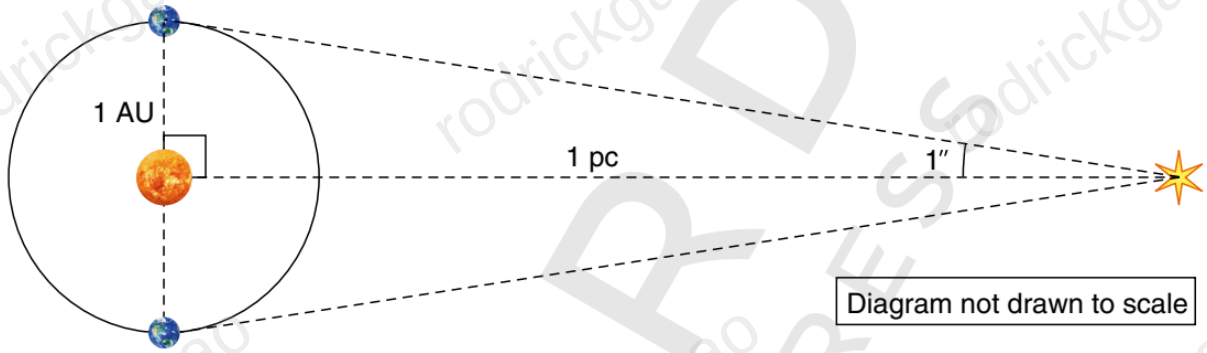


Fig 3.1i Definition of one parsec.

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

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E1.1.6 Retrograde planetary motion

Retrograde motion is a planet's temporary apparent westward motion relative to distant background stars as seen from Earth. Changing viewing geometry and orbital speeds, such as Earth overtaking an outer planet, explain it in the heliocentric model. The planet need not reverse its actual orbit. This differs from daily east-to-west apparent motion.

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E1.1.7 Earth's rotation and daily celestial motion

Earth rotates from west to east, making celestial objects appear to move across the sky from east to west each day. This daily apparent motion differs from a planet's temporary retrograde motion relative to background stars. Identify which reference pattern and timescale the question describes.

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E1.1.8 Phases of Venus and Jupiter's moons

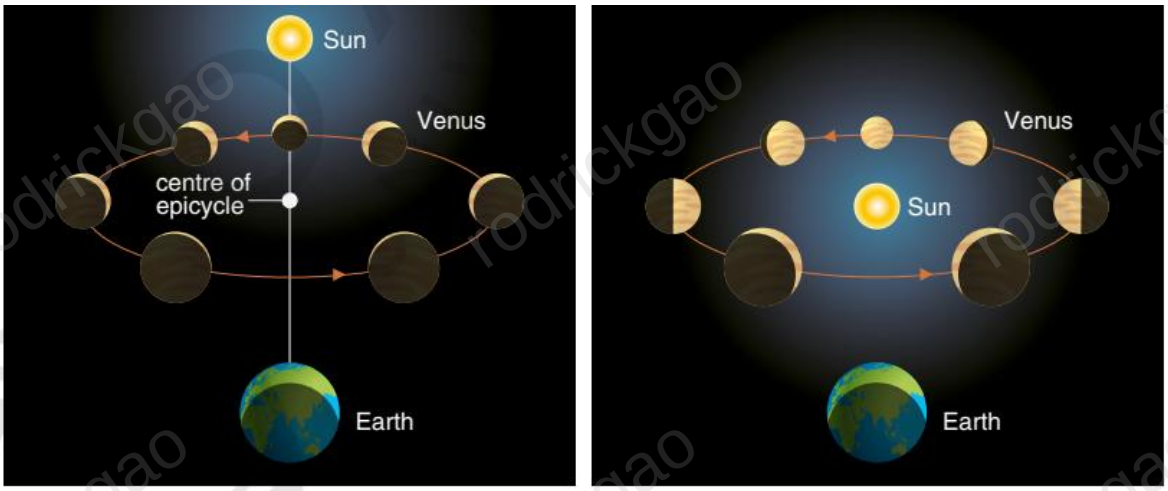
Venus shows a full range of phases, with a larger apparent diameter when nearer and crescent-shaped and a smaller diameter when farther and nearly full. This fits Venus orbiting the Sun but not the Ptolemaic arrangement keeping it between Earth and the Sun. Jupiter's moons directly show that not

everything orbits Earth. Explain which model assumption each observation challenges rather than merely saying it supports heliocentrism.



Fig 1.4h *Different phases of Venus over a period of time. The planet looks bigger when it*

Venus changes apparent diameter as its phase changes.



(i) In the Ptolemaic model, Venue does not go *(ii) In the heliocentric model, Venus shows*

Compare the phases of Venus predicted by the Ptolemaic and heliocentric models.

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rodrickgao